



# HW#5 API Reference

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This document summarizes only the USART and GPIO/LED APIs needed for the starter code.

## USART API

### USART\_GetFlagStatus()

```
FlagStatus USART_GetFlagStatus(USART_TypeDef* USARTx, uint16_t USART_FLAG);
```

Checks whether a USART status flag is set.

Common flags:

```
USART_FLAG_RXNE    /* Receive data register is not empty */  
USART_FLAG_TXE     /* Transmit data register is empty */
```

Example:

```
if (USART_GetFlagStatus(USART3, USART_FLAG_RXNE) == SET) {  
    /* A received character is available. */  
}
```

### USART\_ReceiveData()

```
uint16_t USART_ReceiveData(USART_TypeDef* USARTx);
```

Reads received data from the USART receive data register.

Example:

```
char ch;  
  
ch = (char)USART_ReceiveData(USART3);
```

### USART\_SendData()

```
void USART_SendData(USART_TypeDef* USARTx, uint16_t Data);
```

Sends one data value through USART.

Example:

```
USART_SendData(USART3, (uint16_t)'A');
```

### USART\_Init()

```
void USART_Init(USART_TypeDef* USARTx, USART_InitTypeDef* USART_InitStruct);
```

Initializes a USART peripheral using a `USART_InitTypeDef` configuration.

Common fields:

```
USART_BaudRate  
USART_WordLength  
USART_StopBits  
USART_Parity  
USART_HardwareFlowControl  
USART_Mode
```

Example:

```
USART_InitTypeDef usart_init = {0};  
  
usart_init.USART_BaudRate = 115200;  
usart_init.USART_WordLength = USART_WordLength_8b;  
usart_init.USART_StopBits = USART_StopBits_1;  
usart_init.USART_Parity = USART_Parity_No;  
usart_init.USART_HardwareFlowControl = USART_HardwareFlowControl_None;  
usart_init.USART_Mode = USART_Mode_Rx | USART_Mode_Tx;  
  
USART_Init(USART3, &usart_init);
```

### USART\_Cmd()

```
void USART_Cmd(USART_TypeDef* USARTx, FunctionalState NewState);
```

Enables or disables the USART peripheral.

Example:

```
USART_Cmd(USART3, ENABLE);
```

## GPIO / LED API

### BSP\_LED\_On()

```
void BSP_LED_On(CPU_INT08U led);
```

Turns on the selected board LED.

Example:

```
BSP_LED_On(1);    /* LED1 */  
BSP_LED_On(2);    /* LED2 */  
BSP_LED_On(3);    /* LED3 */
```

### BSP\_LED\_Off()

```
void BSP_LED_Off(CPU_INT08U led);
```

Turns off the selected board LED.

Example:

```
BSP_LED_Off(1);   /* LED1 */  
BSP_LED_Off(2);   /* LED2 */  
BSP_LED_Off(3);   /* LED3 */
```

### BSP\_LED\_Toggle()

```
void BSP_LED_Toggle(CPU_INT08U led);
```

Toggles the selected board LED.

Example:

```
BSP_LED_Toggle(1); /* LED1 */
BSP_LED_Toggle(2); /* LED2 */
BSP_LED_Toggle(3); /* LED3 */
```

## LED Pin Mapping

The starter code assumes the following LED pins:

| LED  | GPIO Port | GPIO Pin |
|------|-----------|----------|
| LED1 | GPIOB     | PB0      |
| LED2 | GPIOB     | PB7      |
| LED3 | GPIOB     | PB14     |

## GPIO / USART Setup API

### RCC\_AHB1PeriphClockCmd()

```
void RCC_AHB1PeriphClockCmd(uint32_t RCC_AHB1Periph, FunctionalState NewState);
```

Enables or disables the AHB1 peripheral clock.

Used for GPIO ports.

Example:

```
RCC_AHB1PeriphClockCmd(RCC_AHB1Periph_GPIOB, ENABLE);
RCC_AHB1PeriphClockCmd(RCC_AHB1Periph_GPIOD, ENABLE);
```

### RCC\_APB1PeriphClockCmd()

```
void RCC_APB1PeriphClockCmd(uint32_t RCC_APB1Periph, FunctionalState NewState);
```

Enables or disables the APB1 peripheral clock.

Used for `USART3`.

Example:

```
RCC_APB1PeriphClockCmd(RCC_APB1Periph_USART3, ENABLE);
```

### `RCC_APB2PeriphClockCmd()`

```
void RCC_APB2PeriphClockCmd(uint32_t RCC_APB2Periph, FunctionalState NewState);
```

Enables or disables the APB2 peripheral clock.

Example:

```
RCC_APB2PeriphClockCmd(RCC_APB2Periph_SYSCFG, ENABLE);
```

### `GPIO_Init()`

```
void GPIO_Init(GPIO_TypeDef* GPIOx, GPIO_InitTypeDef* GPIO_InitStruct);
```

Initializes GPIO pins using a `GPIO_InitTypeDef` configuration.

Common fields:

```
GPIO_Pin  
GPIO_Mode  
GPIO_OType  
GPIO_PuPd  
GPIO_Speed
```

Example for LED pins:

```
GPIO_InitTypeDef gpio_init = {0};  
  
gpio_init.GPIO_Mode  = GPIO_Mode_OUT;  
gpio_init.GPIO_OType = GPIO_OType_PP;  
gpio_init.GPIO_Speed = GPIO_Speed_2MHz;  
gpio_init.GPIO_PuPd  = GPIO_PuPd_NOPULL;  
gpio_init.GPIO_Pin   = GPIO_Pin_0 | GPIO_Pin_7 | GPIO_Pin_14;  
  
GPIO_Init(GPIOB, &gpio_init);
```

Example for USART3 pins:

```

GPIO_InitTypeDef gpio_init = {0};

gpio_init.GPIO_Pin    = GPIO_Pin_8 | GPIO_Pin_9;
gpio_init.GPIO_Mode   = GPIO_Mode_AF;
gpio_init.GPIO_OType  = GPIO_OType_PP;
gpio_init.GPIO_PuPd   = GPIO_PuPd_UP;
gpio_init.GPIO_Speed  = GPIO_Speed_50MHz;

GPIO_Init(GPIOD, &gpio_init);

```

### GPIO\_PinAFConfig()

```

void GPIO_PinAFConfig(GPIO_TypeDef* GPIOx, uint16_t GPIO_PinSource, uint8_t
    GPIO_AF);

```

Connects a GPIO pin to an alternate function.

The starter code uses alternate function mapping for USART3:

```

GPIO_PinAFConfig(GPIOD, GPIO_PinSource8, GPIO_AF_USART3);
GPIO_PinAFConfig(GPIOD, GPIO_PinSource9, GPIO_AF_USART3);

```

## USART3 Pin Mapping

The starter code assumes the following USART3 pins:

| USART Signal | GPIO Port | GPIO Pin |
|--------------|-----------|----------|
| USART3 RX    | GPIOD     | PD8      |
| USART3 TX    | GPIOD     | PD9      |